TUH preprocessing analysis — is the LaBraM protocol right for TUEG and TUAB?

LaBraM, BIOT and CBraMod prepare TUAB with one fixed recipe: keep 23 referential channels, band-pass 0.1–75 Hz, **notch 50 Hz**, resample to 200 Hz, express in µV (`dataset_maker/make_TUAB.py`, walked through step by step in `notebooks/tuab_data_preparation.ipynb`). TUH is recorded in the US, where mains power is **60 Hz**, and the full TUEG corpus mixes sampling rates (250–1000 Hz), equipment and two decades of recordings. Before converting 1.76 TB of TUEG this notebook measures, on a stratified sample of real recordings, what the protocol leaves in the signal and whether any of it could act as a shortcut for brain age.

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The measurements come from `labram/eval/preprocessing_qc.py` (`analyze_recording`), run on a 120 s segment (60–180 s after the start) of each recording.

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187 recordings analysed, 0 errors
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1. The protocol and the variants tested

For a channel $x(t)$ in µV sampled at f_s Hz, the LaBraM protocol computes

$$y = \mathrm{resample}_{200\,\mathrm{Hz}}\left(\mathrm{notch}_{50}\left(\mathrm{bp}_{[0.1,\,75]}(x)\right)\right) \tag{1.1}$$

where $\mathrm{bp}_{[0.1,75]}$ is MNE's zero-phase FIR band-pass (`raw.filter(0.1, 75)`, transition bands chosen automatically: 0.1 Hz below, $\min(\max(0.25 \cdot 75, 2), f_s/2 - 75)$ Hz above), notch_{f_0} is MNE's FIR notch at f_0 Hz (`raw.notch_filter(f0)`), and $\mathrm{resample}_{200}$ is MNE's FFT resampler (`raw.resample(200)`). Table 1.1 lists the variants measured next to it; each changes only the notch or the low-pass of Eq. (1.1) (`labram/eval/preprocessing_qc.py::PROTOCOLS`).

Table 1.1. Protocol variants

One row per variant: notch frequencies (Hz), band-pass edges (Hz), output rate (Hz).

protocol	notch (Hz)	band-pass (Hz)	output rate (Hz)
labram (50 Hz notch)	50	0.1–75	200.00
60 Hz notch	60	0.1–75	200.00
50 + 60 Hz notch	50, 60	0.1–75	200.00
60 Hz notch, 45 Hz low-pass	60	0.1–45	200.00

2. The sample

TUEG recordings were drawn from the corpus index (`dataset_maker/tuh_index.py`), restricted to recordings with all 23 channels, a known age and at least 5 minutes, **one recording per patient**, up to 12 per cell of native sampling rate × recording year. TUAB adds 40 normal and 40 abnormal recordings (one per patient). Table 2.1 gives the counts.

The recording year is the year in the EDF header's `Startdate` field (the date the EEG was recorded, not the patient's age), grouped as ≤2004, 2005–10 and 2011+. Sampling rate and recording year are not independent in TUEG: 400 Hz is almost only ≤2004, 512 Hz and 1000 Hz almost only 2011+, and 256 Hz never appears in 2005–10 (empty cells in Table 2.1). Any property of the equipment therefore also carries the recording date — and through it, as Section 6 shows, the patient's age.

Table 2.1. Sample composition

Recordings per group × native sampling rate (Hz) × recording year (from the EDF header).

group	recording year sfreq	≤2004	2005–10	2011+	total
TUAB	**250**	0	14	62	76
	256	0	0	4	4
	400	0	0	0	0
	512	0	0	0	0
	1000	0	0	0	0
TUEG	**250**	12	12	12	36
	256	12	0	12	24
	400	12	5	1	18
	512	2	0	12	14
	1000	3	0	12	15

Table 2.2 shows the EDF header fields that could say how a recording was already filtered or scaled: the physical unit of the 23 channels and the free-text *prefiltering* field. Every recording is in µV, and the prefiltering field carries no information (`HP:0 LP:0 N:0` , or `-1` placeholders) — the headers cannot tell us whether a hardware notch was applied, so mains interference has to be measured in the signal itself (Section 4).

Table 2.2. EDF header unit and prefiltering fields

Recordings per (unit, prefiltering text, group).

	group	TUAB	TUEG
units	**prefilter**		
uV	**HP:-1.000 Hz LP:-2.0 Hz N:0.0**	0	30
	HP:0.000 Hz LP:0.0 Hz N:0.0	80	77

3. Measures

Spectrum. For a segment $x_c(t)$, $c = 1..C$ channels, the channel-median power spectral density is

$$P(f) = \text{median}_c \;\; \text{Welch}\left(x_c\right)(f) \qquad [\mu\text{V}^2/\text{Hz}] \tag{3.1}$$

with 4 s Hann segments and 50% overlap (0.25 Hz resolution; `preprocessing_qc.py::channel_median_psd`). The median over channels keeps one bad electrode from dominating.

Line-noise prominence. How far a narrow band around f_0 stands above its neighbourhood:

$$\text{LNP}(f_0) = 10 \log_{10} \frac{\text{mean}_{|f-f_0|\le 0.5}\, P(f)}{\text{median}_{2\le|f-f_0|\le 5}\, P(f)} \qquad [\text{dB}] \tag{3.2}$$

(`line_prominence_db`). $\text{LNP} \approx 0$ dB means no line; 10 dB means a line ten times its surroundings; a negative value is a dip, e.g. the hole a notch filter cuts. It is measured at the mains fundamentals (50, 60 Hz), the 2nd/3rd US harmonics (120, 180 Hz), and where harmonics would fold back by aliasing: 180 Hz sampled at 250 or 256 Hz lands at 70 or 76 Hz, and after resampling to 200 Hz, 120 Hz lands at 80 Hz and 180 Hz at 20 Hz.

Mains share of the output. The share of the 0.5–75 Hz power that falls within ±5 Hz of 60 Hz:

$$F_{55\text{–}65} = \frac{\sum_{55 \le f \le 65} P(f)}{\sum_{0.5 \le f \le 75} P(f)} \tag{3.3}$$

(`band_fraction`).

Amplitude and integrity (`amplitude_stats` , on the raw 23-channel segment in µV, T samples):

$$A_{50} = \mathrm{median}_{c,t}\,|x_c(t)|, \qquad A_{99} = \mathrm{quantile}_{0.99,\,c,t}\,|x_c(t)| \qquad [\mu\mathrm{V}] \tag{3.4}$$

$$\mathrm{clip} = \frac{1}{CT} \sum_{c,t} 1\big[\,|x_c(t)| \ge 0.995\, x_{\max}\big] \tag{3.5}$$

where $x_{\max}$ is the EDF physical maximum (±5000 µV for TUH), and

$$\mathrm{flat} = \frac{1}{C} \sum_{c} 1\Big[\min_{k} \mathrm{std}\left(x_c[\text{10 s window } k]\right) < 0.5\,\mu\mathrm{V}\Big] \tag{3.6}$$

is the share of channels that are flat (disconnected or saturated) in some 10 s window.

4. Mains interference in the raw recordings

Figure 4.1 averages $\log_{10} P(f)$ of Eq. (3.1) over the recordings of each sampling-rate group, 40–130 Hz (up to each group's Nyquist frequency). Table 4.1 gives the median LNP of Eq. (3.2) per group and the share of recordings with a clear line ($\mathrm{LNP}_{60} > 6$ dB, a 4× peak).

Figure 4.1. Raw spectrum around the mains frequencies (before any filtering)

(a) TUEG and (b) TUAB: mean over recordings of $\log_{10} P(f)$ (µV²/Hz, Eq. 3.1) of the raw 120 s segment **before any filtering** (native sampling rate, no band-pass, no notch), one line per native sampling rate; dashed lines at 50 and 60 Hz. n per line in Table 2.1. The curves are *means* of log-spectra, so a minority of recordings with strong lines produces a visible peak even where the *median* recording has almost none (Table 4.1).

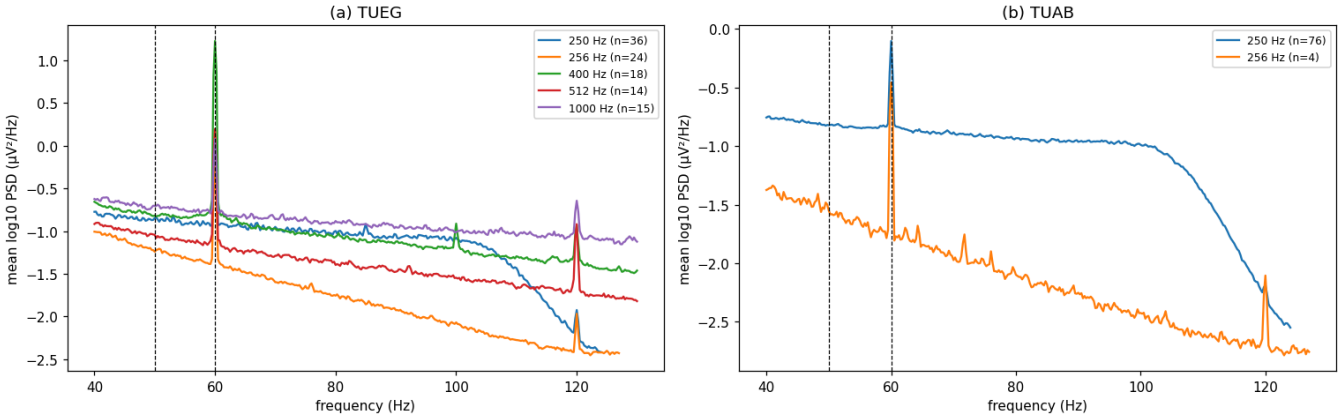


Table 4.1. Line-noise prominence of the raw recordings (before any filtering)

Median LNP (dB, Eq. 3.2) of the unfiltered recordings at each probe frequency per group × native sampling rate; `Hz` = native sampling rate; `60 p75` / `60 p90` = 75th/90th percentile of LNP_{60}; `%>6dB` / `%>20dB` = percentage of recordings with LNP_{60} above 6 / 20 dB; `n` recordings. – where the frequency is above the group's Nyquist limit (half the sampling rate).

group	Hz	50 Hz	60 Hz	120 Hz	180 Hz	70 Hz	76 Hz	60 p75	60 p90	%>6dB	%>20dB	n
TUAB	**250**	0.1	1.9	-0.4	–	0.0	0.1	5.7	12.7	21	8	76
	256	0.2	3.3	0.8	–	-0.3	0.1	11.9	24.7	25	25	4
TUEG	**250**	-0.0	4.5	0.1	–	0.0	0.1	9.1	20.4	36	11	36
	256	-0.1	5.4	1.4	–	0.0	0.1	12.1	18.9	46	8	24
	400	0.0	14.8	1.1	3.6	0.1	-0.0	33.8	40.2	56	44	18
	512	0.0	5.9	3.9	5.3	-0.0	-0.2	14.2	23.8	50	14	14
	1000	0.1	2.8	0.1	0.6	0.2	-0.0	5.3	16.9	20	7	15

Table 4.1 and Figure 4.1 give the first answer. **There is no 50 Hz line anywhere** (median $\mathrm{LNP}_{50} \approx 0$ dB in every group), so the protocol's 50 Hz notch removes no interference — it only cuts a hole in genuine EEG at 50 Hz (Section 5). **A 60 Hz line is present in about half of the recordings** (`%>6dB` per group in Table 4.1; over the whole sample, a third exceed 6 dB), and its strength depends on the equipment: weakest in TUAB and in TUEG's 1000 Hz recordings, strongest in the 400 Hz group (the ≤2004 systems of Table 2.1). The 120 Hz and 180 Hz harmonics are visible where the sampling rate allows them (above 240 / 360 Hz native). The acquisition-alias probes (70, 76 Hz) are flat: the 180 Hz harmonic is not folded into the band.

5. What each protocol leaves at the model input

Every recording was passed through the four protocols of Table 1.1 and Eqs. (3.2)–(3.3) were measured on the 200 Hz output. Figure 5.1 compares the output spectra of the LaBraM protocol and the 60 Hz-notch variant; Table 5.1 summarises all four.

Figure 5.1. Model-input spectrum around 60 Hz after processing, LaBraM vs 60 Hz notch

Mean over recordings of $\log_{10} P(f)$ (µV²/Hz, Eq. 3.1) of the 200 Hz output **after processing**, 40–90 Hz, one line per native sampling rate. Left column: LaBraM protocol (50 Hz notch); right column: the same with a 60 Hz notch instead. (a, b) TUEG (n = 107), (c, d) TUAB (n = 80). Dashed lines at 50, 60 and 75 Hz (low-pass edge).

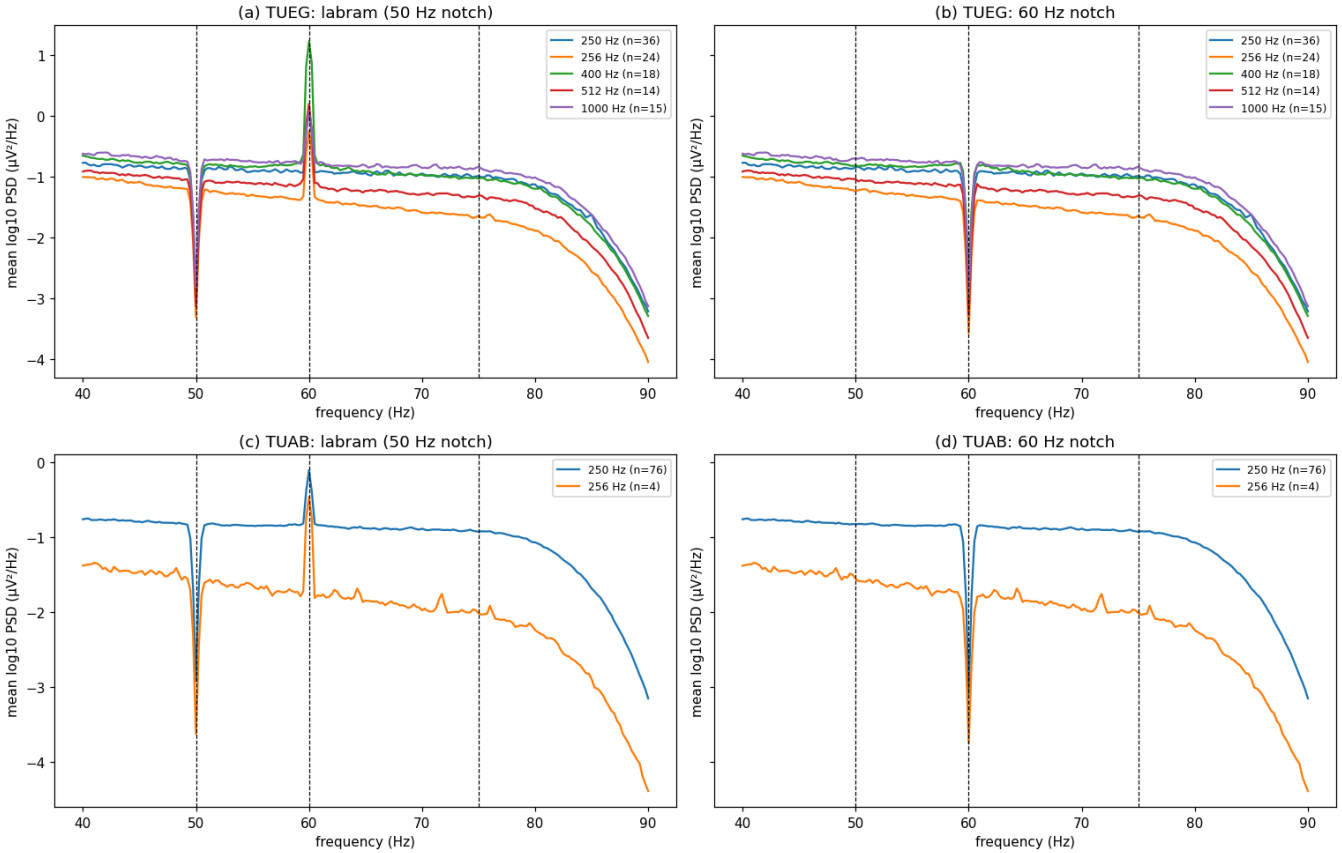


Table 5.1. Line noise left at the model input after processing, per protocol

TUEG + TUAB, all 187 recordings, after processing: median LNP (dB, Eq. 3.2) at 50 and 60 Hz and at the resampling-alias probes 20 and 80 Hz; `share60>6dB` = share of recordings with a 60 Hz line above 6 dB left; `F55-65` = median mains share of the output power (Eq. 3.3, %).

protocol	50 Hz	60 Hz	20 Hz (alias of 180)	80 Hz (alias of 120)	share60>6dB	F55-65 (%)
labram (50 Hz notch)	-5.05	3.13	0.05	0.46	0.33	1.39
60 Hz notch	0.04	-5.35	0.05	0.46	0.01	0.83
50 + 60 Hz notch	-5.05	-5.35	0.05	0.46	0.01	0.83
60 Hz notch, 45 Hz low-pass	-0.42	-5.68	0.06	-0.99	0.01	0.00

Table 5.1 is the core result. The LaBraM protocol passes the 60 Hz line through unchanged (its median LNP_{60} equals the raw value of Table 4.1) and cuts a dip at 50 Hz where there was no interference. A 60 Hz notch removes the line in almost every recording (median LNP_{60} becomes a dip of about −5 dB, the notch itself; `share60>6dB` drops to about 1%, the few recordings whose line is wider than MNE's default notch width of $f_0/200$ Hz). Notching both 50 and 60 Hz only adds the needless 50 Hz dip. Resampling causes no systematic aliasing (median of the 20 and 80 Hz probes ≈0 dB): MNE's FIR low-pass at 75 Hz attenuates 120/180 Hz before the 200 Hz resampler; the 80 Hz probe lies in the low-pass transition band, where the little that remains is already strongly attenuated. A 45 Hz low-pass removes the mains band entirely but also discards 45–75 Hz gamma/EMG content that the pre-trained LaBraM tokenizer was trained on.

6. Is the residual a bias?

A 60 Hz line matters for brain age only if its strength is related to age. Figure 6.1 plots the raw LNP_{60} against age and by recording year; Table 6.1 gives Spearman correlations

$$\rho_S(u, v) = \mathrm{corr}_{\mathrm{Pearson}}\left(\mathrm{rank}(u), \mathrm{rank}(v)\right) \tag{6.1}$$

(`scipy.stats.spearmanr` , two-sided p), between LNP_{60} and age, and between age and the native sampling rate, per group.

Figure 6.1. 60 Hz line strength vs age and recording year

(a) Raw LNP_{60} (dB, Eq. 3.2) against age (years), TUEG (dots, colour = native sampling rate) and TUAB (crosses); (b) raw LNP_{60} by recording year, TUEG. n in Table 2.1.

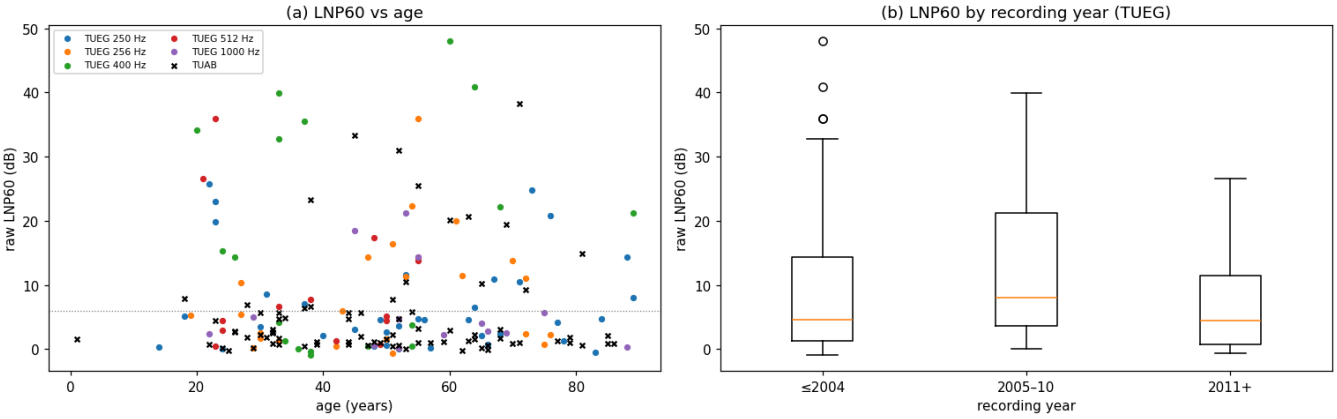


Table 6.1. Correlation of line strength and equipment with age

Spearman ρ_S (Eq. 6.1) and two-sided p per group; `n` recordings.

group	variable vs age	rho_S	p	n
TUAB	**raw LNP60**	-0.026	0.821	80
	native sampling rate	-0.193	0.087	80
	recording year	-0.154	0.173	80
TUEG	**raw LNP60**	0.008	0.936	107
	native sampling rate	-0.148	0.127	107
	recording year	0.098	0.314	107

Table 6.2. Patient age by equipment and recording year across the TUEG corpus

All TUEG recordings of the corpus index with a usable age (not only the sample): mean and median patient age (years) and `n` recordings per native sampling rate and per recording year.

		mean age	median age	n
sampling rate (Hz)	**250**	52.3	55.0	22443
	256	50.5	53.0	34153
	400	46.9	47.0	3057
	512	46.6	46.0	1510
	1000	52.4	54.0	3062
recording year	**≤2004**	45.1	47.0	6089
	2005–10	52.7	55.0	11180
	2011+	51.3	53.0	46643

Within the sample, line strength is **not** directly correlated with age (Table 6.1, $\rho_S \approx 0$ in both groups): the 60 Hz line is not a recording-level age marker. The risk is at group level. Table 6.2 shows on the whole corpus that patients recorded on different equipment generations, and in different recording years, differ in mean age by several years — the 400 Hz and 512 Hz systems and the ≤2004 recordings are the youngest — and Table 4.1 shows that the 400 Hz (≤2004) systems also carry the strongest 60 Hz line. A model can pick up such equipment fingerprints and use them as a proxy for the age mix of the sub-population they come from. Removing the line (Table 5.1) cuts this path at no cost: brain-age models use no physiological signal in a 1 Hz band at 60 Hz. Equipment and recording year should in any case be reported as strata of the evaluation (error per sampling rate and recording year), since other fingerprints than the line remain.

7. Amplitude scale and signal integrity

Table 7.1 gives the amplitude and integrity measures of Eqs. (3.4)–(3.6) per group and, for TUAB, per normal/abnormal label. Figure 7.1 plots A_{50} against age and by sampling rate.

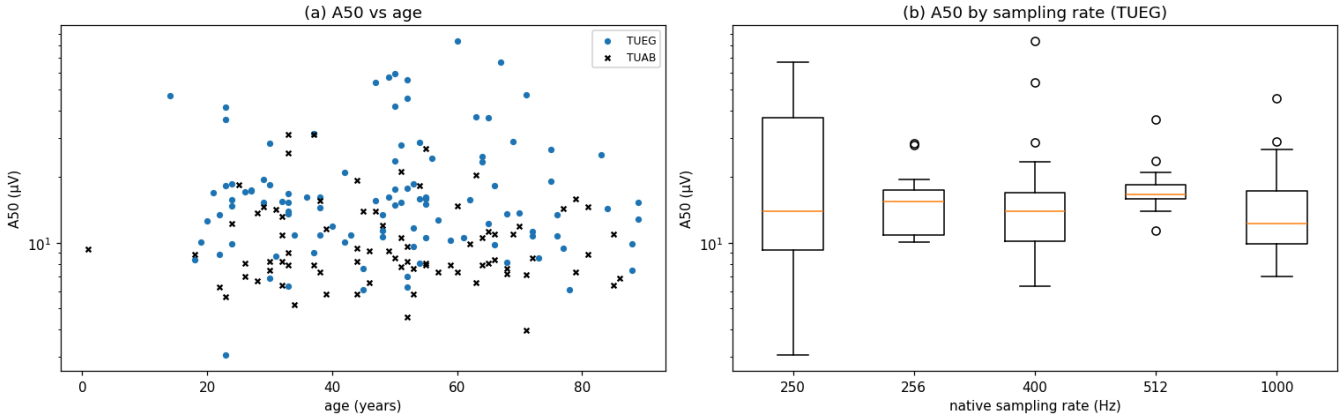
Table 7.1. Amplitude and integrity of the raw signal

Medians over recordings of A_{50}, A_{99} (µV, Eq. 3.4), clip share (Eq. 3.5) and flat-channel share (Eq. 3.6); `n` recordings.

cohort	A50 (µV)	A99 (µV)	clip	flat	n
TUAB abnormal	9.915	77.476	0.000	0.000	40
TUAB normal	8.469	80.910	0.000	0.000	40
TUEG	15.262	124.515	0.000	0.000	107

Figure 7.1. Signal amplitude vs age and sampling rate

(a) A_{50} (µV, Eq. 3.4, log scale) against age (years) for TUEG (dots) and TUAB (crosses); (b) A_{50} by native sampling rate, TUEG. n in Table 2.1.



No recording in the sample clips or has a flat channel (Table 7.1), so integrity filtering is a corpus-level concern (missing channels, very short files; corpus index), not a per-sample one. Amplitude is a different matter: TUEG's typical amplitude is markedly higher than TUAB's (Table 7.1) — TUEG is not curated to routine recordings the way TUAB is — and it varies between sampling-rate groups (Figure 7.1(b)), i.e. between equipment generations. In this sample A_{50} shows no clear trend with age (Figure 7.1(a)), so scale is mainly a corpus/equipment difference here. LaBraM feeds the model $x/100$ (fixed scaling, `labram/train/train_finetune.py`), which keeps both. Per-patch z-scoring (`labram_plus.z_score`, Section 9 of `notebooks/tuab_data_preparation.ipynb`) removes absolute scale entirely; common average reference (`labram_plus.common_average_reference`, on in the brain-age config) removes the reference-dependent part shared across channels. Which of the two to use for TUEG is an empirical question for the training experiments (Section 9).

8. Removing the line without a hole

A notch filter removes a band around 60 Hz whether or not there is interference in it: on a recording without a line it cuts a hole into genuine EEG (the dips in Figure 5.1). Two alternatives remove only the interference.

Sinusoid fitting (MNE `notch_filter(method="spectrum_fit")`, the approach of EEGLAB's CleanLine): in sliding 10 s windows, a 60 Hz sinusoid is estimated by multitaper regression and subtracted,

$$\hat{x}(t) = x(t) - \hat{a}\cos(2\pi\,60\,t) - \hat{b}\sin(2\pi\,60\,t) \qquad (8.1)$$

where $\hat{a}, \hat{b}$ (µV) are the window's fitted amplitudes, kept only when a multitaper F-test finds the line significant.

Spectrum interpolation (Leske & Dalal 2019; `preprocessing_qc.py::spectrum_interpolate`): the short-time Fourier transform $Z(f, \tau)$ with 10 s Hann windows and 50% overlap is modified in the band $|f - 60| \le h$ as

$$Z'(f,\tau) = \left(\mathrm{mean}_{h<|f'-60|\le h+3} |Z(f',\tau)| \right) e^{i\angle Z(f,\tau)}, \qquad |f-60| \le h, \qquad (8.2)$$

i.e. the magnitude in the band is set to the mean magnitude of the 3 Hz on either side and the phase is kept; the inverse transform gives the cleaned signal. h = 1.5 Hz. Away from the band the transform is an exact round trip.

Both are compared with the 60 Hz notch on two subsets of the sample: recordings **with a strong line** (raw $\mathrm{LNP}_{60} > 10$ dB), where the question is how much of the line is left, and **clean** recordings (raw $|\mathrm{LNP}_{60}| < 1$ dB), where the question is how deep a hole the method cuts. Table 8.1 gives the numbers, Figure 8.1 the spectra.

Table 8.1. Line left and hole cut by each line-removal method (after processing)

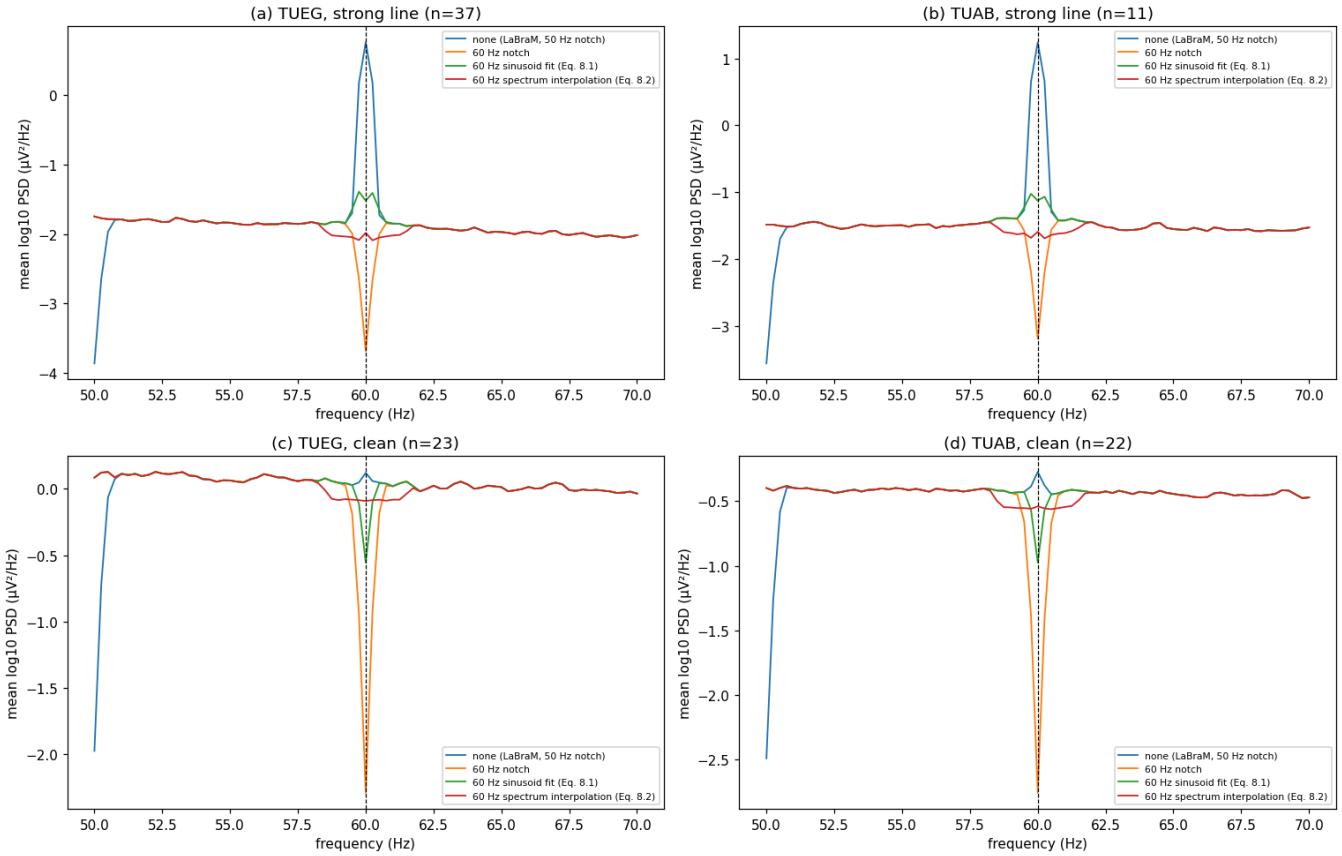
LNP_{60} (dB, Eq. 3.2) of the 200 Hz output after band-pass, the line-removal method and resampling. Strong-line recordings (raw $\mathrm{LNP}_{60} > 10$ dB): median and maximum line left (lower is better, 0 dB = no line). Clean recordings (raw $|\mathrm{LNP}_{60}| < 1$ dB): median LNP_{60}, where a negative value is the hole the method cuts (closer to 0 is better). `n` recordings per subset; TUEG + TUAB.

method	strong: median left (dB)	strong: max left (dB)	clean: median (dB)	clean: min (dB)	n strong / clean
none (LaBraM, 50 Hz notch)	20.0	48.0	0.4	-1.0	48 / 45
60 Hz notch	-4.8	7.4	-5.6	-6.9	48 / 45
60 Hz sinusoid fit (Eq. 8.1)	2.0	26.6	-1.3	-2.7	48 / 45
60 Hz spectrum interpolation (Eq. 8.2)	-1.6	1.5	-1.3	-1.8	48 / 45

Figure 8.1. Spectrum around 60 Hz after each line-removal method

Mean over recordings of $\log_{10} P(f)$ (µV²/Hz, Eq. 3.1) of the 200 Hz output after processing, 50–70 Hz, one line per method of Table 8.1: (a) TUEG and (b) TUAB recordings with a strong

line (raw $\mathrm{LNP}_{60} > 10$ dB); (c) TUEG and (d) TUAB clean recordings (raw $|\mathrm{LNP}_{60}| < 1$ dB). n in each panel title. A good method is flat at 60 Hz in all four panels: no peak in (a, b), no dip in (c, d).



Spectrum interpolation removes the line best and cuts no meaningful hole. On the strong-line recordings it leaves no line at all (Table 8.1: median about −1.6 dB, worst case 1.5 dB — better than the notch's worst case of 7.4 dB, whose fixed narrow band misses lines that are broader than it), and on clean recordings it lowers the 60 Hz band by only about 1 dB, against a hole of about 5.6 dB from the notch (Figure 8.1(c, d)). Sinusoid fitting sits in between: on clean recordings it still cuts a narrow dip exactly at 60 Hz (shallower than the notch's, Figure 8.1(c, d)), and on strong-line recordings it misses part of the line (worst case above 25 dB, Figure 8.1(a, b)) — TUH's line drifts in amplitude within the 10 s fitting window, so a single stationary sinusoid does not describe it. After interpolation the 60 Hz region is nearly flat in all four panels of Figure 8.1: only a shallow plateau of about 1 dB remains across the ±1.5 Hz band, while the notch leaves a deep dip in every cohort.

9. Conclusions and the recommended TUH protocol

1. **Mains: remove 60 Hz, not 50 Hz — by spectrum interpolation.** No recording has 50 Hz interference (Table 4.1); about half have a 60 Hz line whose strength depends on the equipment generation, which is itself tied to recording year and patient age at group level (Tables 2.1, 6.2), though not to age recording by recording (Table 6.1). The

LaBraM protocol passes this line to the model unchanged (Table 5.1, Figure 5.1(a, c)). Spectrum interpolation (Eq. 8.2, h = 1.5 Hz) removes it from every recording without the hole a notch cuts (Table 8.1, Figure 8.1); a 60 Hz notch is the fallback. To be confirmed by a TUAB training comparison (notch vs interpolation) before the TUEG conversion.
2. **Keep the 0.1–75 Hz band-pass and 200 Hz resampling.** They cause no aliasing at any native rate from 250 to 1000 Hz (alias probes in Tables 4.1 and 5.1) and match what the pre-trained LaBraM weights expect. A 45 Hz low-pass is not needed once the line is removed.
3. **Units are consistent** (all µV, Table 2.2); the header prefiltering field is uninformative, so per-recording QC must come from the signal.
4. **Amplitude differs by corpus and equipment** (Table 7.1, Figure 7.1); keep CAR, and compare per-patch z-scoring against fixed scaling in the first TUEG experiments.
5. **Consistency:** TUAB must be preprocessed with exactly the same recipe as TUEG when the two are mixed. For the TUAB benchmark (comparison with published numbers) the original 50 Hz recipe stays available.

Table 9.1 summarises the recommendation.

Table 9.1. LaBraM protocol vs the recommended TUH protocol

One row per preprocessing step; `evidence` names the table or figure that supports the change.

step	LaBraM (TUAB papers)	recommended for TUH	evidence
channels	23 referential, exact set required	23 referential, missing ones zero-filled + channel_mask	corpus index (23% of TUEG recordings miss ≥1 channel)
band-pass	0.1–75 Hz	0.1–75 Hz (unchanged)	Tables 4.1, 5.1
mains removal	50 Hz notch	60 Hz spectrum interpolation (±1.5 Hz); 60 Hz notch as fallback	Tables 4.1, 5.1, 6.2, 8.1; Figures 5.1, 8.1
resample	200 Hz	200 Hz (unchanged)	Table 5.1 (no aliasing)
units / scale	µV, model input x/100	µV; CAR on; z-score vs fixed scaling tested	Table 7.1, Figure 7.1
TUAB	as published	re-made with the TUH recipe when mixed with TUEG; published recipe kept for the benchmark	Section 9, item 5